

# Network\_CheatSheet

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SHUBHAM

Cheatsheet for DevOps Engineers

## Networking Basics:

| Command                                                   | Description                                          |
|-----------------------------------------------------------|------------------------------------------------------|
| <code>ifconfig</code>                                     | Show or configure network interfaces                 |
| <code>ifconfig eth0 up</code>                             | Activate eth0 network interface                      |
| <code>ip addr</code>                                      | Show or change IP addresses                          |
| <code>ip addr show</code>                                 | Show network interfaces and associated IP addresses  |
| <code>ip addr add 192.168.1.1/24 dev eth0</code>          | Add IP address with subnet mask to network interface |
| <code>ip link</code>                                      | Show or configure network interfaces                 |
| <code>ip link show</code>                                 | Show network interface information                   |
| <code>ip link set eth0 down</code>                        | Deactivate eth0 network interface                    |
| <code>ip route</code>                                     | Show or change routing table                         |
| <code>ip route show</code>                                | Show current routing table                           |
| <code>ip route add default via &lt;192.168.1.1&gt;</code> | Add default gateway on 192.168.1.1.                  |



# Advanced Networking:

| Command                                                                                                | Description                                   |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <code>route</code>                                                                                     | Show or configure routing table               |
| <code>route -n</code>                                                                                  | Show routing table (numeric addresses)        |
| <code>route flush</code>                                                                               | Removes all routes                            |
| <code>route add default gw &lt;192.168.1.1&gt;</code>                                                  | Add default route to routing table.           |
| <code>arp</code>                                                                                       | Show or change ARP cache                      |
| <code>arp -a</code>                                                                                    | Prints arp table                              |
| <code>arp -n</code>                                                                                    | Show ARP cache (numeric addresses)            |
| <code>arp -a -d</code>                                                                                 | Deletes all arp table entries                 |
| <code>arp -d &lt;192.168.1.100&gt;</code>                                                              | Remove 192.168.1.100 from the ARP cache       |
| <code>arp -s</code>                                                                                    | Adds entry in arp table                       |
| <code>iwconfig</code>                                                                                  | Manage wireless network interfaces            |
| <code>iwconfig wlan0</code>                                                                            | Show informationn for wlan0 network interface |
| <code>curl</code><br><code>wget</code>                                                                 | Download files from the internet              |
| <code>curl -o &lt;link&gt;</code><br><code>wget &lt;link&gt;</code><br><code>area2 &lt;link&gt;</code> | Download file and save in current directory   |

## Network Monitoring:

| Command                           | Description                                                                   |
|-----------------------------------|-------------------------------------------------------------------------------|
| <i>netstat</i>                    | <i>Network statistics</i>                                                     |
| <i>netstat -tuln</i>              | <i>Show active network sockets</i>                                            |
| <i>netstat -r</i>                 | <i>Show kernel routing table</i>                                              |
| <i>ss</i>                         | <i>Socket statistics</i>                                                      |
| <i>ss -tuln</i>                   | <i>Show active network sockets</i>                                            |
| <i>ss -i</i>                      | <i>Show network interface packet statistics</i>                               |
| <i>iftop</i>                      | <i>Real time bandwidth usage</i>                                              |
| <i>iftop -n</i>                   | <i>Real time bandwidth usage (numeric addresses)</i>                          |
| <i>iftop -i eth0</i>              | <i>Real time bandwidth usage for eth0 network interface</i>                   |
| <i>tcpdump</i>                    | <i>Network packet analyzer</i>                                                |
| <i>tcpdump -i &lt;eth0&gt;</i>    | <i>Show network traffic on eth0 network interface</i>                         |
| <i>tcpdump -n &lt;port 80&gt;</i> | <i>Show network traffic on port 80 (numeric addresses)</i>                    |
| <i>nc / netcat /ncat</i>          | <i>Provides the ability to read and write data across network connections</i> |
| <i>hping</i>                      | <i>Analyzes and exchanges TCP/IP packets with a remote host.</i>              |
| <i>speedometer</i>                | <i>Displays bandwidth usage in real-time</i>                                  |
| <i>vnstat</i>                     | <i>Logs and shows time-based network traffic stats.</i>                       |
| <i>socat</i>                      | <i>Transfers data between two bidirectional byte streams</i>                  |



DNS and Host Resolution:

| Command                                   | Description        |
|-------------------------------------------|--------------------|
| <code>dig &lt;example.com&gt;</code>      | Perform DNS lookup |
| <code>nslookup &lt;example.com&gt;</code> | Query DNS servers  |
| <code>host &lt;example.com&gt;</code>     | Perform DNS lookup |
| <code>hostname</code>                     | Display hostname   |
| <code>hostname &lt;myhost&gt;</code>      | Change hostname    |

DNS and Host Resolution:

| Command                                      | Description                               |
|----------------------------------------------|-------------------------------------------|
| <code>ping &lt;example.com&gt;</code>        | Send ICMP echo requests                   |
| <code>tracert &lt;example.com&gt;</code>     | Trace route to destination                |
| <code>tracertpath &lt;example.com&gt;</code> | Simplified tracert                        |
| <code>mtr &lt;example.com&gt;</code>         | Combines ping and tracert functionalities |
| <code>whois &lt;example.com&gt;</code>       | Lookup information for IP or domain.      |

Connectivity:

| Command                    | Description                                                   |
|----------------------------|---------------------------------------------------------------|
| <code>ifconfig eth0</code> | Check if ethernet cable is plugged in eth0                    |
| <code>iperf</code>         | Tests network performance within two systems                  |
| <code>bwmon</code>         | Monitors current bandwidth for multiple network interfaces    |
| <code>telnet</code>        | Connects to a remote system via Telnet over a TCP/IP network. |



| Command | Description                                                       |
|---------|-------------------------------------------------------------------|
| w       | Displays information about currently logged-in users              |
| mail    | Sends and receives email using the command line.                  |
| iw      | Displays and configures wireless network interfaces               |
| ngrep   | Displays and filters network packet data on a given regex pattern |

SSH & Remote Access:

| Command | Description                                                            |
|---------|------------------------------------------------------------------------|
| ssh     | Securely connects to a remote system using the SSH protocol            |
| scp     | Copies files securely between client and server using the SSH protocol |
| sftp    | Securely transfers files between hosts using the SFTP protocol         |

Security:

| Command   | Description                                                                      |
|-----------|----------------------------------------------------------------------------------|
| iptables  | Firewall utility that manages packet filtering and NAT.                          |
| snort     | Intrusion detection system that analyzes network traffic for suspicious activity |
| wireshark | Captures and analyzes network traffic in a formatted text                        |
| ufw       | Manages system firewall and adds/deletes/modifies/resets packet filtering rule   |